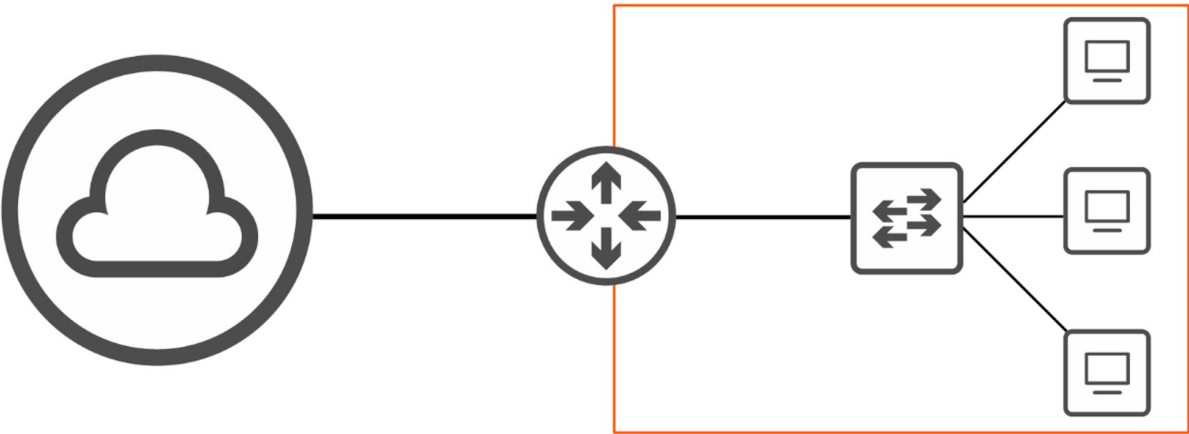


# Day 5 - Ethernet LAN Switching (pt 1)

Thursday, August 20, 2026 3:28 PM

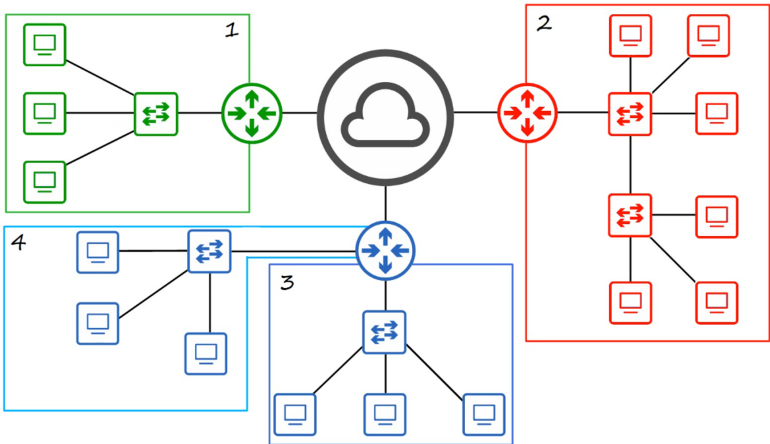
## Ethernet LAN Switching



### LAN's

- A LAN is a network contained in a relatively small area.
- Routers are used to connect separate LAN's

## Local Area Networks (LANs)



Data

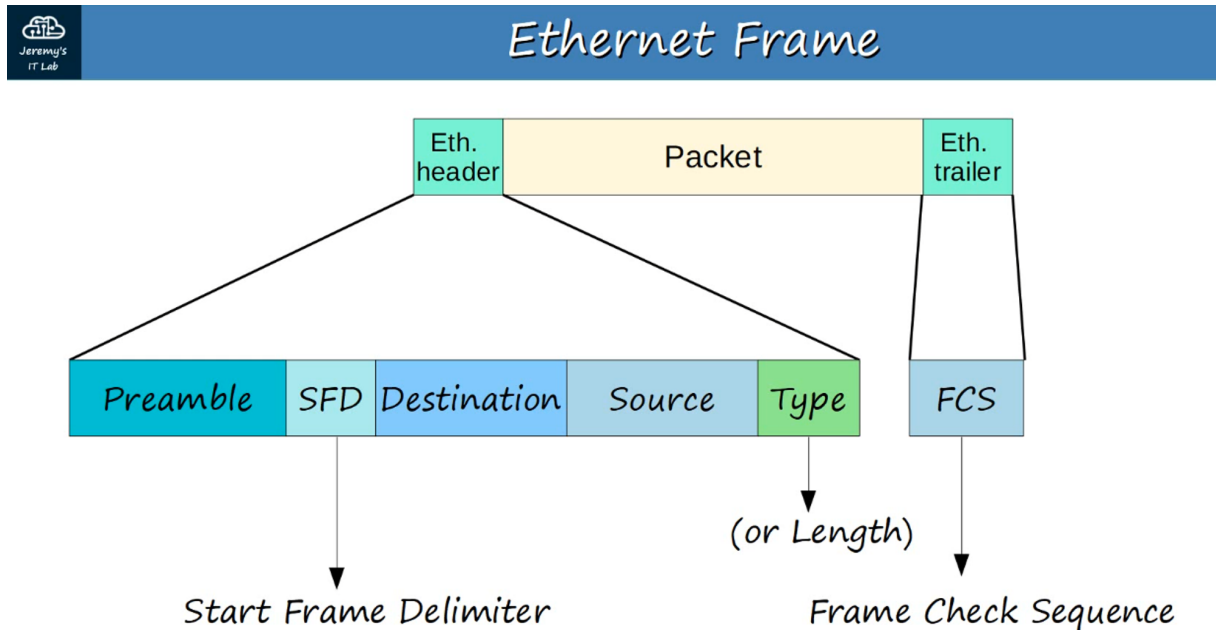
Segment

Packet

Protocol Data Units (PDUs)



An ETHERNET FRAME looks like:



Ethernet Trailer --- PACKET --- Ethernet Header

The Ethernet Header contains 5 Fields:

Preamble -- SFD -- Destination -- Source -- Type 7 bytes -- 1 byte -- 6 bytes -- 6 bytes -- 2 bytes

#### PREAMBLE:

- Length: 7 bytes (56 bits)
- Alternating 1's and 0's
- 10101010 \* 7x
- Allows devices to synchronize their receiver clocks

#### SFD : 'Start Frame Delimiter'

- Length: 1 byte(8 bits)
- 10101011
- Marks end of the PREAMBLE and beginning of rest of frame.

#### DESTINATION AND SOURCE

- Layer 2 Address
- Indicates the devices sending / receiving the frame
- MAC = 'Media Access Control'
- = 6 byte (48-bit) address of the physical device

#### TYPE / LENGTH

- 2 bytes (16-bit) field
- A value of 1500 or less in this field indicates the LENGTH of the encapsulated packet (in bytes)
- A value of 1536 or greater in this field indicates the TYPE of the encapsulated packet and length is determined via other methods.
- IPv4 = 0x0800 (hexadecimal) = 2048 in decimal

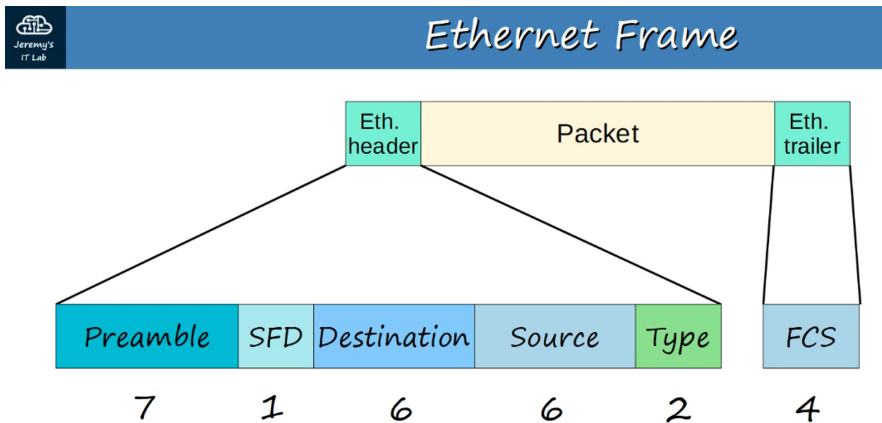
- IPv6 = 0x86DD (hexadecimal) = 34525 in decimal
- Layer 3 protocol used in the encapsulated Packet, which is almost always Internet Protocol (IP) version 4 or version 6.

The ETHERNET TRAILER contains:

FCS

- 'FRAME CHECK SEQUENCE'
- 4 bytes (32 bits) in length
- Detects corrupted data by running a 'CRC' algorithm over the received data
- CRC = "Cyclic Redundancy Check"

Altogether the ETHERNET FRAME = 26 bytes (header + trailer)



= 26 bytes (header + trailer)

### MAC ADDRESS (48 bits long)

- 6-bytes (48-bits) physical address assigned to the device when it is made.
- AKA 'Burned-In Address' (BIA)
- Is globally unique
- First 3 bytes are the OUI (Organizationally Unique Identifier) which is assigned to the company making the device
- The last 3 bytes are unique to the device itself
- Written as 12 hexadecimal characters

Example:

E8:BA:70 // 11:28:74 OUI // Unique Device ID

### HEXADECIMAL

Hexadecimal

| DEC. | HEX. | DEC. | HEX. | DEC. | HEX. | DEC. | HEX. |
|------|------|------|------|------|------|------|------|
| 0    | 0    | 8    | 8    | 16   | 10   | 24   | 18   |
| 1    | 1    | 9    | 9    | 17   | 11   | 25   | 19   |
| 2    | 2    | 10   | A    | 18   | 12   | 26   | 1A   |

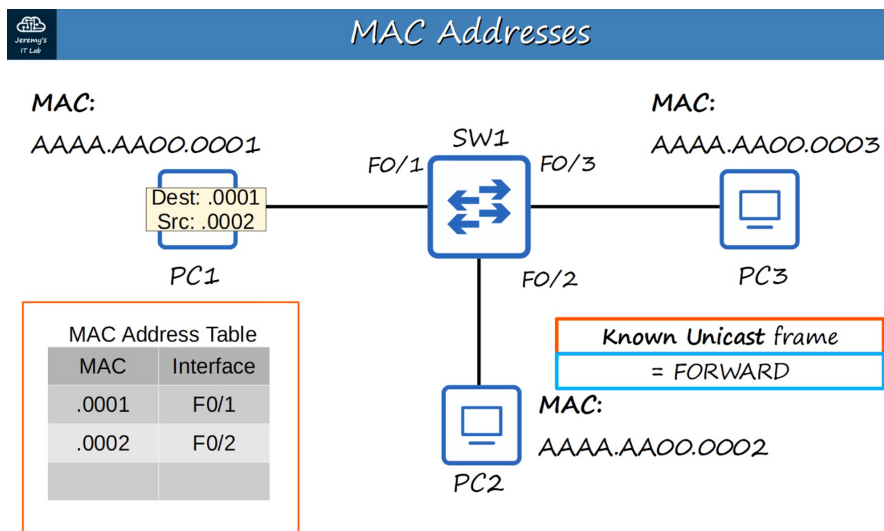
|   |   |    |   |    |    |    |    |
|---|---|----|---|----|----|----|----|
| 2 | 2 | 10 | A | 18 | 12 | 26 | 1A |
| 3 | 3 | 11 | B | 19 | 13 | 27 | 1B |
| 4 | 4 | 12 | C | 20 | 14 | 28 | 1C |
| 5 | 5 | 13 | D | 21 | 15 | 29 | 1D |
| 6 | 6 | 14 | E | 22 | 16 | 30 | 1E |
| 7 | 7 | 15 | F | 23 | 17 | 31 | 1F |

## INTERFACE NAMES

F0/1, F0/2, F0/3... F stands for "Fast Ethernet" or 100 Mbps interfaces.

## MAC ADDRESS TABLE

Each Switch stores a DYNAMICALLY LEARNED MAC ADDRESS TABLE, using the SOURCE MAC ADDRESS of frames it receives.



When a Switch doesn't know the DESTINATION MAC ADDRESS of a frame (**UNKNOWN UNICAST FRAME**), it is forced to FLOOD the frame - Forward the frame out of ALL it's interfaces, except the one it received the packet from.

When a **KNOWN Unicast Frame** is known (MAC Address is recognized by the entry in the MAC ADDRESS TABLE), the frame is FORWARDED like normal.

